

The Pile: An 800GB Dataset of Diverse Text for Language Modeling

Leo Gao	Stella Biderman	Sid Black	Laurence Golding
Travis Hoppe	Charles Foster	Jason Phang	Horace He
Anish Thite	Noa Nabeshima	Shawn Presser	Connor Leahy

EleutherAI
contact@eleuther.ai

Abstract

Recent work has demonstrated that increased training dataset diversity improves general cross-domain knowledge and downstream generalization capability for large-scale language models. With this in mind, we present *the Pile*: an 825 GiB English text corpus targeted at training large-scale language models. The Pile is constructed from 22 diverse high-quality subsets—both existing and newly constructed—many of which derive from academic or professional sources. Our evaluation of the untuned performance of GPT-2 and GPT-3 on the Pile shows that these models struggle on many of its components, such as academic writing. Conversely, models trained on the Pile improve significantly over both Raw CC and CC-100 on all components of the Pile, while improving performance on downstream evaluations. Through an in-depth exploratory analysis, we document potentially concerning aspects of the data for prospective users. We make publicly available the code used in its construction.¹

versity leads to better downstream generalization capability (Rosset, 2019). Additionally, large-scale language models have been shown to effectively acquire knowledge in a novel domain with only relatively small amounts of training data from that domain (Rosset, 2019; Brown et al., 2020; Carlini et al., 2020). These results suggest that by mixing together a large number of smaller, high quality, diverse datasets, we can improve the general cross-domain knowledge and downstream generalization capabilities of the model compared to models trained on only a handful of data sources.

To address this need, we introduce the Pile: a 825.18 GiB English text dataset designed for training large scale language models. The Pile is composed of 22 diverse and high-quality datasets, including both established natural language processing datasets and several newly introduced ones. In addition to its utility in training large language models, the Pile can also serve as a broad-coverage benchmark for cross-domain knowledge and generalization ability of language models.

1 Introduction

Recent breakthroughs in general-purpose language modeling have demonstrated the effectiveness of training massive models on large text corpora for downstream applications (Radford et al., 2019; Shueybi et al., 2019; Raffel et al., 2019; Rosset, 2019; Brown et al., 2020; Lepikhin et al., 2020). As the field continues to scale up language model training, the demand for high-quality massive text data will continue to grow (Kaplan et al., 2020).

The growing need for data in language modeling has caused most existing large-scale language models to turn to the Common Crawl for most or all of their data (Brown et al., 2020; Raffel et al., 2019). While training on the Common Crawl has been effective, recent work has shown that dataset di-

We introduce new datasets derived from the following sources: PubMed Central, ArXiv, GitHub, the FreeLaw Project, Stack Exchange, the US Patent and Trademark Office, PubMed, Ubuntu IRC, HackerNews, YouTube, PhilPapers, and NIH ExPorter. We also introduce OpenWebText2 and BookCorpus2, which are extensions of the original OpenWebText (Gokaslan and Cohen, 2019) and BookCorpus (Zhu et al., 2015; Kobayashi, 2018) datasets, respectively.

In addition, we incorporate several existing high-quality datasets: Books3 (Presser, 2020), Project Gutenberg (PG-19) (Rae et al., 2019), OpenSubtitles (Tiedemann, 2016), English Wikipedia, DM Mathematics (Saxton et al., 2019), EuroParl (Koehn, 2005), and the Enron Emails corpus (Klimt and Yang, 2004). To supplement these, we also in-

¹<https://pile.eleuther.ai/>

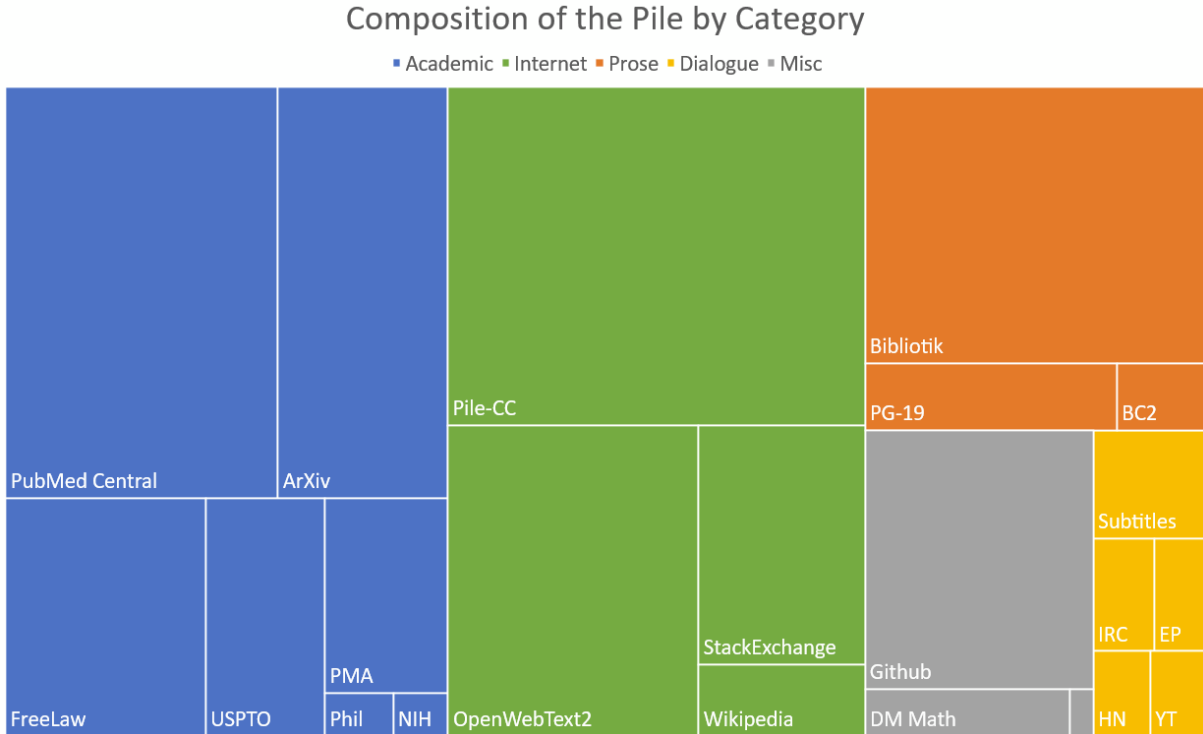


Figure 1: Treemap of Pile components by effective size.

roduce a new filtered subset of Common Crawl, Pile-CC, with improved extraction quality.

Through our analyses, we confirm that the Pile is significantly distinct from pure Common Crawl data. Additionally, our evaluations show that the existing GPT-2 and GPT-3 models perform poorly on many components of the Pile, and that models trained on the Pile significantly outperform both raw and filtered Common Crawl models. To complement the performance evaluations, we also perform an exploratory analysis of the text within the Pile to provide a detailed picture of the data. We hope that our extensive documentation of the construction and characteristics of the Pile will help researchers make informed decisions about potential downstream applications.

Finally, we make publicly available the preprocessing code for the constituent datasets of the Pile and the code for constructing alternative versions². In the interest of reproducibility, we also document all processing performed on each dataset (and the Pile as a whole) in as much detail as possible. For further details about the processing of each dataset, see Section 2 and Appendix C.

²<https://github.com/EleutherAI/the-pile>

1.1 Contributions

The core contributions of this paper are:

1. The introduction of a 825.18 GiB english-language dataset for language modeling combining 22 diverse sources.
2. The introduction of 14 new language modeling datasets, which we expect to be of independent interest to researchers.
3. Evaluations demonstrating significant improvements across many domains by GPT-2-sized models trained on this new dataset, compared to training on CC-100 and raw Common Crawl.
4. The investigation and documentation of this dataset, which we hope will better inform researchers about how to use it as well as motivate them to undertake similar investigations of their own data.

2 The Pile Datasets

The Pile is composed of 22 constituent sub-datasets, as shown in Table 1. Following Brown et al. (2020), we increase the weights of higher quality components, with certain high-quality datasets such as Wikipedia being seen up to 3 times (“epochs”) for

Component	Raw Size	Weight	Epochs	Effective Size	Mean Document Size
Pile-CC	227.12 GiB	18.11%	1.0	227.12 GiB	4.33 KiB
PubMed Central	90.27 GiB	14.40%	2.0	180.55 GiB	30.55 KiB
Books3 [†]	100.96 GiB	12.07%	1.5	151.44 GiB	538.36 KiB
OpenWebText2	62.77 GiB	10.01%	2.0	125.54 GiB	3.85 KiB
ArXiv	56.21 GiB	8.96%	2.0	112.42 GiB	46.61 KiB
Github	95.16 GiB	7.59%	1.0	95.16 GiB	5.25 KiB
FreeLaw	51.15 GiB	6.12%	1.5	76.73 GiB	15.06 KiB
Stack Exchange	32.20 GiB	5.13%	2.0	64.39 GiB	2.16 KiB
USPTO Backgrounds	22.90 GiB	3.65%	2.0	45.81 GiB	4.08 KiB
PubMed Abstracts	19.26 GiB	3.07%	2.0	38.53 GiB	1.30 KiB
Gutenberg (PG-19) [†]	10.88 GiB	2.17%	2.5	27.19 GiB	398.73 KiB
OpenSubtitles [†]	12.98 GiB	1.55%	1.5	19.47 GiB	30.48 KiB
Wikipedia (en) [†]	6.38 GiB	1.53%	3.0	19.13 GiB	1.11 KiB
DM Mathematics [†]	7.75 GiB	1.24%	2.0	15.49 GiB	8.00 KiB
Ubuntu IRC	5.52 GiB	0.88%	2.0	11.03 GiB	545.48 KiB
BookCorpus2	6.30 GiB	0.75%	1.5	9.45 GiB	369.87 KiB
EuroParl [†]	4.59 GiB	0.73%	2.0	9.17 GiB	68.87 KiB
HackerNews	3.90 GiB	0.62%	2.0	7.80 GiB	4.92 KiB
YoutubeSubtitles	3.73 GiB	0.60%	2.0	7.47 GiB	22.55 KiB
PhilPapers	2.38 GiB	0.38%	2.0	4.76 GiB	73.37 KiB
NIH ExPorter	1.89 GiB	0.30%	2.0	3.79 GiB	2.11 KiB
Enron Emails [†]	0.88 GiB	0.14%	2.0	1.76 GiB	1.78 KiB
The Pile	825.18 GiB			1254.20 GiB	5.91 KiB

Table 1: Overview of datasets in the Pile before creating the held out sets. Raw Size is the size before any up- or down-sampling. Weight is the percentage of bytes in the final dataset occupied by each dataset. Epochs is the number of passes over each constituent dataset during a full epoch over the Pile. Effective Size is the approximate number of bytes in the Pile occupied by each dataset. Datasets marked with a [†] are used with minimal preprocessing from prior work.

each full epoch over the Pile. Detailed information about the construction of each dataset is available in Appendix C.

2.1 Pile-CC

Common Crawl is a collection of website crawls from 2008 onwards, including raw web pages, metadata and text extractions. Due to the raw nature of the dataset, Common Crawl has the advantage of including text from diverse domains, but at the cost of varying quality data. Due to this, use of Common Crawl typically necessitates well-designed extraction and filtering. Our Common Crawl-based dataset, Pile-CC, uses just-Text (Endrédy and Novák, 2013) on Web Archive files (raw HTTP responses including page HTML) for extraction, which yields higher quality output than directly using the WET files (extracted plain-text).

2.2 PubMed Central

PubMed Central (PMC) is a subset of the PubMed online repository for biomedical articles run by the United States of America’s National Center for Biotechnology Information (NCBI), providing open, full-text access to nearly five million publications. Most publications indexed by PMC are recent, and their inclusion is mandated for all NIH funded research starting from 2008 by the NIH Public Access Policy. We included PMC in the hopes that it will benefit potential downstream applications to the medical domain.

2.3 Books3

Books3 is a dataset of books derived from a copy of the contents of the Bibliotik private tracker made available by Shawn Presser (Presser, 2020). Bibliotik consists of a mix of fiction and nonfiction books and is almost an order of magnitude

larger than our next largest book dataset (BookCorpus2). We included Bibliotik because books are invaluable for long-range context modeling research and coherent storytelling.

2.4 OpenWebText2

OpenWebText2 (OWT2) is a generalized web scrape dataset inspired by WebText (Radford et al., 2019) and OpenWebTextCorpus (Gokaslan and Cohen, 2019). Similar to the original WebText, we use net upvotes on Reddit submissions as a proxy for outgoing link quality. OpenWebText2 includes more recent content from Reddit submissions up until 2020, content from multiple languages, document metadata, multiple dataset versions, and open source replication code. We included OWT2 as a high quality general purpose dataset.

2.5 ArXiv

ArXiv is a preprint server for research papers that has operated since 1991. As shown in fig. 10, arXiv papers are predominantly in the fields of Math, Computer Science, and Physics. We included arXiv in the hopes that it will be a source of high quality text and math knowledge, and benefit potential downstream applications to research in these areas. ArXiv papers are written in LaTeX, a common typesetting language for mathematics, computer science, physics, and some adjacent fields. Training a language model to be able to generate papers written in LaTeX could be a huge boon to the research community.

2.6 GitHub

GitHub is a large corpus of open-source code repositories. Motivated by the ability of GPT-3 (Brown et al., 2020) to generate plausible code completions despite its training data not containing any explicitly gathered code datasets, we included GitHub in the hopes that it would enable better downstream performance on code-related tasks.

2.7 FreeLaw

The Free Law Project is a US-registered non-profit that provides access to and analytical tools for academic studies in the legal realm. CourtListener,³ part of the Free Law Project, provides bulk downloads for millions of legal opinions from federal and state courts. While the full dataset provides multiple modalities of legal proceedings, including dockets, bibliographic information on judges,

³<https://www.courtlistener.com/>

and other metadata, we focused specifically on court opinions due to an abundance of full-text entries. This data is entirely within the public domain.

2.8 Stack Exchange

The Stack Exchange Data Dump⁴ contains an anonymized set of all user-contributed content on the Stack Exchange network, a popular collection of websites centered around user-contributed questions and answers. It is one of the largest publicly available repositories of question-answer pairs, and covers a wide range of subjects—from programming, to gardening, to Buddhism. We included Stack Exchange in the hopes that it will improve the question answering capabilities of downstream models on diverse domains.

2.9 USPTO Backgrounds

USPTO Backgrounds is a dataset of background sections from patents granted by the United States Patent and Trademark Office, derived from its published bulk archives⁵. A typical patent background lays out the general context of the invention, gives an overview of the technical field, and sets up the framing of the problem space. We included USPTO Backgrounds because it contains a large volume of technical writing on applied subjects, aimed at a non-technical audience.

2.10 Wikipedia (English)

Wikipedia is a standard source of high-quality text for language modeling. In addition to being a source of high quality, clean English text, it is also valuable as it is written in expository prose, and spans many domains.

2.11 PubMed Abstracts

PubMed Abstracts consists of the abstracts from 30 million publications in PubMed, the online repository for biomedical articles run by the National Library of Medicine. While the PMC (see Section 2.2) provides full-text access, the subset of coverage is significantly limited and biased towards recent publications. PubMed also incorporates MEDLINE, which expands the coverage of biomedical abstracts from 1946 to present day.

⁴<https://archive.org/details/stackexchange>

⁵<https://bulkdata.uspto.gov/>

2.12 Project Gutenberg

Project Gutenberg is a dataset of classic Western literature. The specific Project Gutenberg derived dataset we used, PG-19, consists of Project Gutenberg books from before 1919 (Rae et al., 2019), which represent distinct styles from the more modern Books3 and BookCorpus. Additionally, the PG-19 dataset is already being used for long-distance context modeling.

2.13 OpenSubtitles

The OpenSubtitles dataset is an English language dataset of subtitles from movies and television shows gathered by Tiedemann (2016). Subtitles provide an important source of natural dialog, as well as an understanding of fictional formats other than prose, which may prove useful for creative writing generation tasks such as screenwriting, speechwriting, and interactive storytelling.

2.14 DeepMind Mathematics

The DeepMind Mathematics dataset consists of a collection of mathematical problems from topics such as algebra, arithmetic, calculus, number theory, and probability, formatted as natural language prompts (Saxton et al., 2019). One major weakness of large language models has been performance on mathematical tasks (Brown et al., 2020), which may be due in part to a lack of math problems in the training set. By explicitly including a dataset of mathematical problems, we hope to improve the mathematical ability of language models trained on the Pile.

2.15 BookCorpus2

BookCorpus2 is an expanded version of the original BookCorpus (Zhu et al., 2015), a widely used language modeling corpus consisting of books written by “as of yet unpublished authors.” BookCorpus is therefore unlikely to have significant overlap with Project Gutenberg and Books3, which consist of published books. BookCorpus is also commonly used as dataset for training language models (Radford et al., 2018; Devlin et al., 2019; Liu et al., 2019).

2.16 Ubuntu IRC

The Ubuntu IRC dataset is derived from the publicly available chatlogs⁶ of all Ubuntu-related channels on the Freenode IRC chat server. Chatlog data

⁶<https://irclogs.ubuntu.com/>

provides an opportunity to model real-time human interactions, which feature a level of spontaneity not typically found in other modes of social media.

2.17 EuroParl

EuroParl (Koehn, 2005) is a multilingual parallel corpus originally introduced for machine translation but which has also seen use in several other fields of NLP (Groves and Way, 2006; Van Halteren, 2008; Ciobanu et al., 2017). We use the most current version at time of writing, which consists of the proceedings of the European Parliament in 21 European languages from 1996 until 2012.

2.18 YouTube Subtitles

The YouTube Subtitles dataset is a parallel corpus of text gathered from human generated closed-captions on YouTube. In addition to providing multilingual data, Youtube Subtitles is also a source of educational content, popular culture, and natural dialog.

2.19 PhilPapers

The PhilPapers⁷ dataset consists of open-access philosophy publications from an international database maintained by the Center for Digital Philosophy at the University of Western Ontario. We included PhilPapers because it spans a wide body of abstract, conceptual discourse, and its articles contain high quality academic writing.

2.20 NIH Grant Abstracts: ExPORTER

The NIH Grant abstracts provides a bulk-data repository for awarded applications through the ExPORTER⁸ service covering the fiscal years 1985-present. We included the dataset because it contains examples of high-quality scientific writing.

2.21 Hacker News

Hacker News⁹ is a link aggregator operated by Y Combinator, a startup incubator and investment fund. Users submit articles defined as “anything that gratifies one’s intellectual curiosity,” but submitted articles tend to focus on topics in computer science and entrepreneurship. Users can comment on submitted stories, resulting in comment trees discussing and critiquing submitted stories. We

⁷<https://philpapers.org/>

⁸<https://exporter.nih.gov/>

⁹<https://news.ycombinator.com>

scrape, parse, and include these comment trees since we believe they provide high quality dialogue and debate on niche topics.

2.22 Enron Emails

The Enron Emails dataset (Klimt and Yang, 2004) is a valuable corpus commonly used for research about the usage patterns of email. We included Enron Emails to aid in understanding the modality of email communications, which is typically not found in any of our other datasets.

3 Benchmarking Language Models with the Pile

While the Pile was conceived as a training dataset for large-scale language models, its coverage of multiple disparate domains makes it also suitable as an evaluation dataset. In this section, we describe how the Pile can be used as a broad-coverage dataset for benchmarking language models.

3.1 Benchmarking Guidelines

The Pile is provided as train, validation, and testing splits. The validation and testing components each contain 0.1% of the data, sampled uniformly at random. While this is a far smaller percentage than most datasets, the sheer size of the dataset results in over 1 GiB of validation and testing data each. We highlight that while we have made efforts to deduplicate documents within the Pile (See: Section D.2), it is still possible that some documents are duplicated across the train/validation/test splits.

Our preferred metric is bits per UTF-8 encoded byte (BPB). Bits per byte is preferred over bits per character or perplexity when using Pile as a metric due to its invariance to different tokenization schemes and the ambiguity of measuring characters in Unicode. To compute bits per byte from a given negative log likelihood loss ℓ , we compute $\text{BPB} = (L_T/L_B) \log_2(e^\ell) = (L_T/L_B)\ell/\ln(2)$, where L_T is the length of the dataset in tokens and L_B is the length of the dataset in UTF-8 encoded bytes. We find that L_T/L_B is 0.29335 GPT-2-tokens/byte across the Pile; dataset-specific values of L_T/L_B can be found in Table 7.

3.2 Test Perplexity with GPT-2 and GPT-3

We compute the test perplexity of the constituent datasets of the Pile using GPT-2 (Radford et al.,

2019) and GPT-3 (Brown et al., 2020), shown in Figure 2. We use all available versions of GPT-2, and all four versions of GPT-3 available via the OpenAI API. Because of the cost associated with using the OpenAI API, we evaluate on one-tenth of the respective test sets for most of the constituent datasets. We report the perplexity converted to bits per UTF-8 encoded byte (BPB). Importantly, we compute perplexity by evaluating each document independently within each dataset, as opposed to concatenating all documents as is common practice for computing perplexity on large corpora.

Full details of the perplexity computation can be found in Appendix E.2.

Unsurprisingly, larger language models generally attain lower perplexity compared to smaller models. Recent work has shown an increased focus on the empirical scaling laws of language models (Kaplan et al., 2020; Henighan et al., 2020). As such, we investigate the scaling law for the GPT-2 and GPT-3 families of models on perplexity evaluation on the Pile. The scaling law relation for the GPT-3 family of models is shown in Figure 2.¹⁰ The line of best fit shown in the figure has a coefficient of -0.1674 and an intercept of 2.5516.

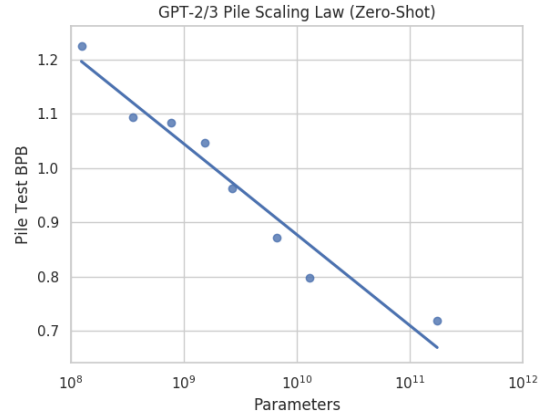


Figure 2: Scaling law for performance of GPT-2/3 models. ‘Zero-shot’ refers to the fact that none of the models have been fine-tuned on data from the Pile.

Interestingly, while GPT-2 and GPT-3 were not trained on the Pile, there still appears to be a clear scaling law without diminishing returns. We hypothesize that this is due to the inherent generalization capability of these models. We leave a more

¹⁰While the sizes of GPT-3 models on the OpenAI API have not been publicized, we assume here that *ada*, *babbage*, *curie* and *davinci* models correspond to 2.7B, 6.7B, 13B and 175B parameter models respectively.

rigorous analysis of zero-shot scaling laws to future work.

3.3 Relative Componentwise GPT-3 Pile Performance

Determining which components GPT-3 underperforms on provides information about which Pile components are most dissimilar to the distribution of text (web pages and books) that GPT-3 was trained on. These components would thus make especially good candidates for supplementing GPT-3 training data. These results are also valuable for determining which types of datasets to emphasize for future iterations of the Pile.

Due to the difference in entropy of different datasets, directly comparing perplexity of GPT-3 on different Pile components is not an accurate indication of relative performance. Ideally we would train a GPT-3 model from scratch on the Pile and compare the difference in loss per dataset with that of the original GPT-3. Because of resource constraints, we instead use a GPT-2 model trained from scratch on the Pile (see Section 4) to construct a proxy measure. To construct our proxy, we first measure the improvement from the GPT-2-Pile model to GPT-3 on each component. Then, we normalize our results by setting the change on OpenWebText2 to be zero. This computation is shown in the equation below:

$$\Delta_{\text{set}} = (L_{\text{set}}^{\text{GPT3}} - L_{\text{owt2}}^{\text{GPT3}}) - (L_{\text{set}}^{\text{GPT2Pile}} - L_{\text{owt2}}^{\text{GPT2Pile}})$$

Since GPT2-Pile was trained on both OWT2 and the dataset we are evaluating, we expect the second term in Δ_{set} to reflect the difference in the intrinsic difficulty of the two datasets. Thus the total value of Δ_{set} reflects how much harder the dataset we are evaluating was for GPT-3 than OWT2, minus the relative difficulty of the two tasks. As GPT-3 was trained on data very similar to OWT2, this gives us a proxy for how much better GPT-3 would do if it were trained on the Pile.

The results are shown in Figure 3. As a sanity check, we observe that datasets that are contained in, or are extremely similar to, GPT-3’s training set (Books3, Wikipedia (en), Pile-CC and Project Gutenberg) score close to zero on our metric.

GPT-3 appears to perform poorly on datasets pertaining to research or academic writing like PubMed Central, PubMed Abstracts, and ArXiv; domain-specific datasets like FreeLaw, HackerNews, and USPTO Backgrounds; and on datasets containing predominantly text distinct from natural language, like GitHub and DM Mathematics. In addition, the majority of datasets see less of an improvement than OpenWebText2. As such, we expect a GPT-3 sized model trained on Pile to perform significantly better on research related tasks, software tasks, and symbol manipulation tasks than the base model. Additionally, this experiment provides evidence that the majority of Pile components are not redundant with the predominantly web-based GPT-3 training data.

We note that this metric is only a proxy for similarity, and that it could be confounded by dataset specific scaling effects. Although our results largely accord with expectations, there are some puzzling results, like the datasets on which GPT-3 outperformed GPT-2 Pile. We hypothesize that GPT-3 learns to be so good at these datasets that training on them explicitly does not notably benefit the model’s performance. We leave a more rigorous analysis of these effects for future work.

4 Evaluation

To confirm the effectiveness of the Pile for improving language modeling quality, we train architecturally-identical 1.3 billion parameter models based on those in Brown et al. (2020) on different datasets and evaluate on the WikiText and LAMBADA tasks as benchmarks of language modeling ability. We also report results on the Pile as a measure of more cross-domain generalization.

4.1 Methodology

To ensure a fair comparison across datasets of different sizes, we decontaminate any instances of the evaluation sets using the same 13-gram overlap filtering as in Brown et al. (2020) and downsample to 40GB to control for dataset size. As we control for dataset size, we emphasize that our evaluation is generous to CC-100 (en), which is about 1/3 the size of the Pile in reality.

We compare the following datasets: the Pile, the En-

Component	GPT-2				GPT-3			
	small	medium	large	xl	ada	babbage	curie	davinci
Pile-CC	1.0878	0.9992	0.9582	0.9355	0.9212	0.8483	0.7849	0.7070
PubMed Central	1.0759	0.9788	0.9334	0.9044	0.8633	0.7792	0.7150	0.6544
Books3	1.1959	1.1063	1.0588	1.0287	0.9778	0.9005	0.8284	0.7052
OpenWebText2	1.1111	1.0073	0.9539	0.9171	0.8727	0.7921	0.7199	0.6242
ArXiv	1.3548	1.2305	1.1778	1.1381	1.0304	0.9259	0.8453	0.7702
Github	1.7912	1.3180	1.7909	1.6486	0.8761	0.7335	0.6415	0.5635
FreeLaw	1.0512	0.9321	0.9017	0.8747	0.8226	0.7381	0.6667	0.6006
Stack Exchange	1.2981	1.1075	1.0806	1.0504	1.0096	0.8839	0.8004	0.7321
USPTO Backgrounds	0.8288	0.7564	0.7202	0.6969	0.6799	0.6230	0.5752	0.5280
PubMed Abstracts	0.9524	0.8579	0.8108	0.7810	0.8130	0.7382	0.6773	0.6201
Gutenberg (PG-19)	1.2655	1.1140	1.0820	1.0829	0.9776	0.8749	0.7930	0.7115
OpenSubtitles	1.2465	1.1657	1.1324	1.1129	1.1116	1.0488	0.9875	0.9130
Wikipedia (en)	1.1285	1.0213	0.9795	0.9655	0.8757	0.7863	0.7047	0.5953
DM Mathematics	2.6911	2.5448	2.4833	2.4377	2.3249	2.2015	2.1067	2.0228
Ubuntu IRC	1.8466	1.7187	1.6427	1.6024	1.3139	1.1968	1.0995	0.9915
BookCorpus2	1.1295	1.0498	1.0061	0.9783	0.9754	0.9041	0.8435	0.7788
EuroParl	2.3177	2.0204	1.8770	1.7650	1.0475	0.9363	0.8415	0.7519
HackerNews	1.4433	1.2794	1.3143	1.3361	1.1736	1.0875	1.0175	0.9457
YoutubeSubtitles	2.0387	1.8412	1.7355	1.6694	1.3407	1.1876	1.0639	0.9469
PhilPapers	1.3203	1.2163	1.1688	1.1327	1.0362	0.9530	0.8802	0.8059
NIH ExPorter	0.9099	0.8323	0.7946	0.7694	0.7974	0.7326	0.6784	0.6239
Enron Emails	1.5888	1.4119	1.4535	1.4222	1.2634	1.1685	1.0990	1.0201
The Pile	1.2253	1.0928	1.0828	1.0468	0.9631	0.8718	0.7980	0.7177

Table 2: Test perplexity of the Pile using GPT-2 and GPT-3, converted to bits per UTF-8 encoded byte (BPB). Evaluation is performed on one-tenth of the test data of the Pile, on a per-document basis. **Bold** indicates the best-performing model in each row.

glish component of the CC-100 dataset¹¹ (Wenzek et al., 2019; Conneau et al., 2020), and a sample of raw CC WET files filtered for English-only.

4.2 Results

On traditional language modeling benchmarks, the Pile improves significantly on WikiText and shows negligible changes in LAMBADA. However, models trained on Pile improve significantly over both Raw CC and CC-100 on all components of the Pile, as shown in Table 4. This indicates that models trained on the Pile have greater cross-domain generalization capabilities without compromising performance on traditional benchmarks.

The magnitude of improvement over CC-100 per set is shown in Figure 4. Unsurprisingly, there is almost no improvement on Pile-CC. However, the model trained on the Pile performs significantly better than either of the other models on academic datasets such as ArXiv, Pubmed Central, FreeLaw, and PhilPapers. It also improves signifi-

cantly on programming-related datasets like Github and StackExchange, on EuroParl, due to the lack of multilingual text in either other dataset, and on DM Mathematics, indicating a significant improvement in mathematical ability.

Surprisingly, raw Common Crawl performs better on the Pile BPB than CC-100, despite losing by a significant margin on LAMBADA and WikiText. We hypothesize that this is due to the perplexity based filtering used in CC-100, where a language model is trained on Wikipedia and all data with a perplexity too high or too low is discarded. This effectively discards any data too similar to or too different from Wikipedia, which severely limits the diversity of the collected data. This result suggests that future work using Common Crawl should take caution with filtering to preserve its diversity.

5 Structural Statistics

In this section, we cover the Structural Statistics of the dataset, which provide more coarse-grained and statistical information about the Pile. In Sec-

¹¹The data was obtained from <http://data.statmt.org/cc-100/>.

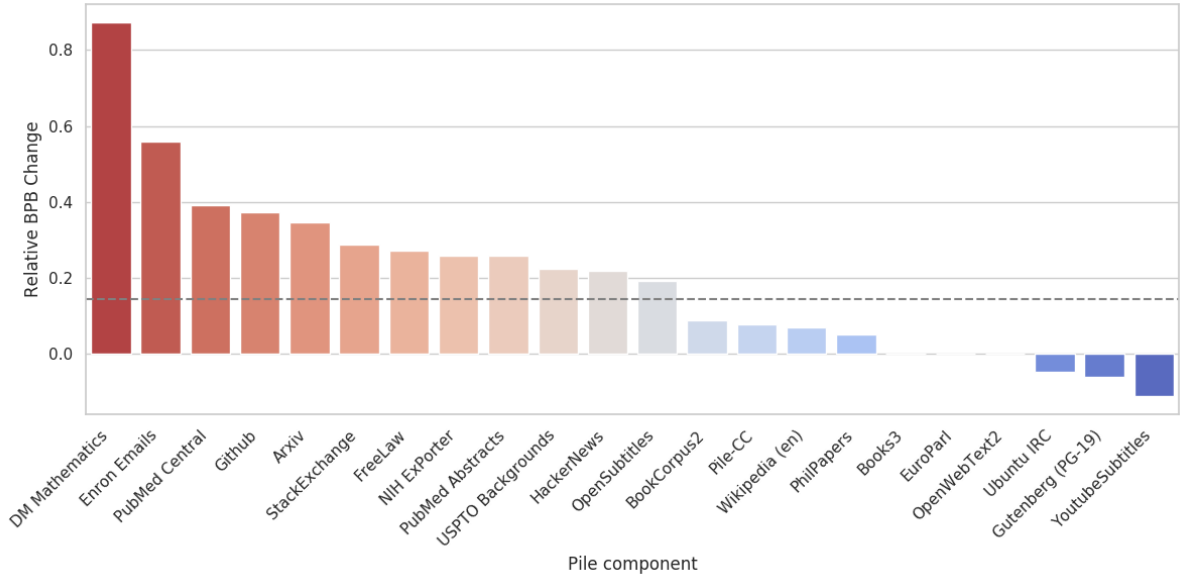


Figure 3: Change in BPB from GPT-2 trained on Pile to GPT-3 zero-shot, relative to OpenWebText2 BPB change. Dotted line indicates overall Pile change. Lower indicates better relative performance by GPT-3.

	Dataset Size	Pile (val) (BPB)	Pile (test) (BPB)	WikiText (PPL)	LAMBADA (PPL)	LAMBADA (ACC)
The Pile	825 GiB	0.9281	0.9433	5.59	12.78	50.1
CC-100 (en)	300 GiB	1.3143	1.3293	8.27	11.78	49.7
Raw CC	45927 GiB [†]	1.1180	1.1275	11.75	19.84	43.8

Table 3: Size-controlled evaluation results. Each dataset is deduplicated against all evaluation metrics and subsampled to approximately 40GB to control for the effects of dataset size. For LAMBADA, we use the variant of the data introduced in Radford et al. (2019) and only evaluate the perplexity on the final token rather than the final word. For WikiText, we report the perplexity per GPT-2 token. [†] indicates that the size is an estimate.

tion 6, we provide a closer investigation and documentation of the textual content within the Pile datasets.

5.1 Document Lengths and Tokenization

Each dataset consists of a large number of documents. We analyze the distribution of document lengths, as well as the number of bytes-per-token using the GPT-2 tokenizer in order to put our ablations in context.

While the majority of documents in the Pile are short, there is a long tail of very long documents (Figure 5).

Since the GPT-2 BPE tokenizer is trained on WebText, the mean bytes per token is also a very rough indicator of how syntactically different each Pile component is from WebText. For instance, datasets like NIH ExPorter, OpenWebText2 and Books3

consist largely of ordinary text in a similar distribution to WebText, which is reflected in a greater number of bytes per token. On the other hand, many of the sets with the lowest bytes per token are those which consist in large part of non-text content (Github, ArXiv, Stack Exchange, and DM Mathematics) or languages other than English (EuroParl).

5.2 Language and Dialects

While only 13% of the world’s population speaks English, the vast majority of NLP research is done on English. For the Pile, we took a similar approach to the dataset used by Brown et al. (2020) and focused predominantly on English, while also not explicitly filtering out other languages when collecting our own data. When evaluating a multilingual dataset, our main criteria for inclusion was whether the English component of the dataset merited inclusion alone. We plan to create a fully multi-

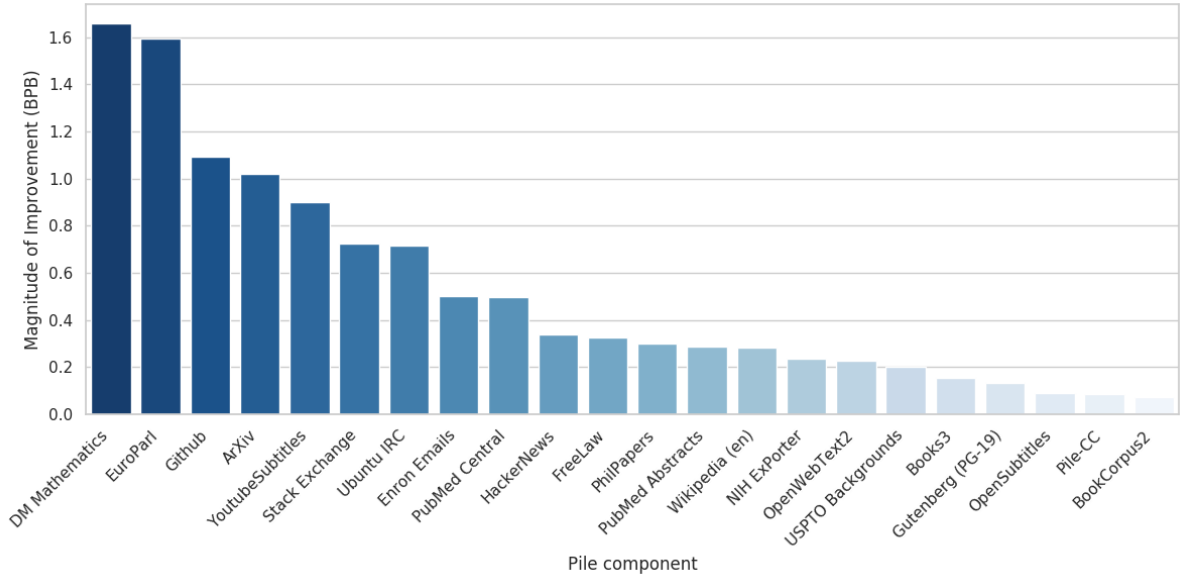


Figure 4: Magnitude of BPB improvement of Pile model over CC-100 model on each test set.

Dataset	The Pile	CC-100 (en)	Raw CC (en)
Pile-CC	0.9989	1.0873	1.0287
PubMed Central	0.6332	1.1311	0.9120
Books3	1.0734	1.2264	1.1366
OpenWebText2	0.9938	1.2222	1.0732
ArXiv	0.7945	1.8159	1.2642
Github	0.5597	1.6509	0.9301
FreeLaw	0.6978	1.0221	0.9468
Stack Exchange	0.8152	1.5414	1.1292
USPTO Backgrounds	0.6731	0.8772	0.8455
PubMed Abstracts	0.7313	1.0193	0.9718
Gutenberg (PG-19)	1.1426	1.2780	1.2235
OpenSubtitles	1.0909	1.1827	1.2139
Wikipedia (en)	0.8961	1.1807	1.0252
DM Mathematics	1.5206	3.1774	2.6229
Ubuntu IRC	1.4085	2.1243	1.5691
BookCorpus2	1.0613	1.1346	1.0914
EuroParl	1.1202	2.7141	1.4917
HackerNews	1.0968	1.4352	1.2305
YoutubeSubtitles	1.4269	2.3287	1.5607
PhilPapers	1.1256	1.4269	1.2090
NIH ExPorter	0.7347	0.9713	0.9225
Enron Emails	0.8301	1.3300	1.0483

Table 4: Breakdown of BPB on Pile heldout test set. Columns indicate the dataset each model is trained on; rows indicate the evaluation dataset. **Bold** indicates the best performing model in each row.

lingual expansion of the Pile as future work.

Using fasttext (Suárez et al., 2019a), we determine that the Pile is 97.4% English. We note that due to issues with language identification, particularly with rare languages Caswell et al. (2020), this methodology provides only a rough estimate for English content and no reliable conclusions for low-resource languages can be drawn.

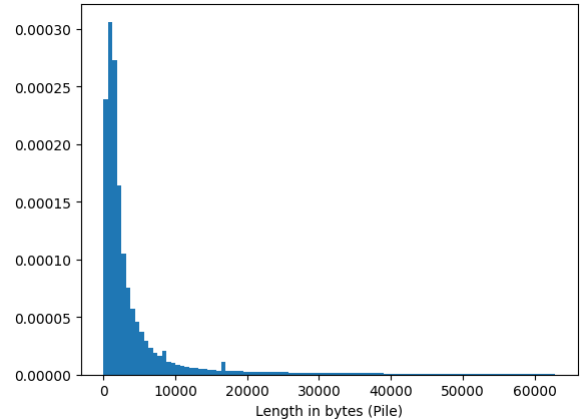


Figure 5: Distribution of document lengths in Pile. The highest 1 percentile of document length are considered to be outliers and excluded from this plot.

6 Investigating and Documenting the Datasets

As the scale of machine learning research has grown, scrutiny has been placed on the ever larger datasets that models are trained on (Prabhu and Birhane, 2020; Biderman and Scheirer, 2020)

While this issue has been raised within AI ethics and bias research (Hovy and Spruit, 2016; Hutchinson et al., 2020; Blodgett et al., 2020), it has not been a focal point of concern within the language modeling community. Despite the proliferation of work exploring and documenting issues with datasets (Gebru et al., 2018; Bender and Friedman,

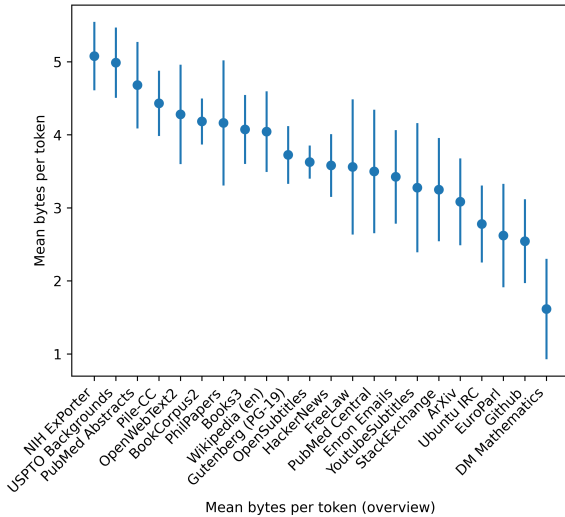


Figure 6: Mean bytes per GPT-2-token for each dataset in the Pile. Error bars indicate standard deviation.

2018; Jo and Gebru, 2020), no dataset intended to train massive language models has been seriously documented by its creators¹². Therefore, our analyses serve two goals: to address ethical concerns about the Pile, and to promote and normalize the practice of engaging with the AI ethics literature.

Natural language processing technologies are widely applicable and can be used in extremely different contexts. What is and is not appropriate data to train on can therefore vary wildly with the application context. In our view, the best approach is to *document* rather than *eliminate* potentially concerning aspects of datasets¹³, particularly since the purpose of the Pile is to train general-purpose language models. The primary goal of our documentation, therefore, is to empower NLP researchers to make informed decisions.

6.1 Documenting Methods

To document the Pile, we chose to implement two frameworks that have been proposed by methodologists and ethics researchers. The first, the datasheets methodology (Gebru et al., 2018), is a general purpose methodology that is recommended by several methodologists (Raji and Yang, 2019; Biderman and Scheirer, 2020) and appears to be used more frequently by practitioners than alterna-

¹²Brown et al. (2020) discusses ethical issues surrounding their *model*, but do not discuss those surrounding the training dataset itself.

¹³That said, we did exclude several datasets, see Appendix B for details.

tives (Seck et al., 2018; Costa-jussà et al., 2020; Thieme et al., 2020). The second, the data statements methodology (Bender and Friedman, 2018), was proposed specifically for natural language processing and has been well received by the NLP community. Our datasheet and data statement will be featured in the GitHub repository where the code for the Pile is stored and will also be available as separate documents on arXiv (Biderman et al., 2021; Biderman, 2021).

In addition to the datasheet and data statement, there is additional information that may be helpful to people training language models that these documents do not cover. In the rest of this section we investigate and document in greater detail some of this additional contextual information.

6.2 Topical Distribution

In order to better understand the specific subject matter covered by the Pile, we performed a topic modeling analysis on its components. Using Gensim (Rehurek et al., 2011), we trained 16-topic Latent Dirichlet Allocation (Blei et al., 2003) models on each component of the validation set of the Pile concurrently, in an online fashion (Hoffman et al., 2010). We filtered the Pile for English only for this analysis. Afterwards, we computed the perplexity of the Common Crawl-derived (Pile-CC) topic model on the document sets of the other components. In this way, we provide a rough measure of the degree to which parts of the Pile contain topics not well covered within Common Crawl.

In Figure 7, these cross-component perplexities are shown, with a vertical line indicating the perplexity of the Pile-CC topic model evaluated on the documents of OpenWebText2. This component was chosen as a baseline of comparison for similar reasons as in the previous evaluation: it is derived in a similar manner (filtered crawls of the open web) as the Common Crawl, and thus is expected to contain a similar distribution of topics. Although Pile-CC is somewhat diverse in its content, several of the Pile’s other components deviate from it strongly in their topical focus, as evidenced by higher perplexity on Github, PhilPapers, and EuroParl.

We also documented the topical clusters inferred from our LDA models for each component, which we provide in Appendix C. As expected, though the larger CC-derived component itself represents a diversity of content—including politics, education,

sports and entertainment—the content clusters it misses become apparent when compared qualitatively to other components of the Pile. Notably, the data modes covering programming, logic, physics, and legal knowledge appear largely absent.

6.3 Pejorative Content

Due to the wide diversity in origins, it is possible for the Pile to contain pejorative, sexually explicit, or otherwise objectionable content. As this content may not be desirable for some use cases, we break down profanity on a per-dataset level.

We used the `profanity-checker` Python package (Zhou, 2019). This package includes a “toxicity model” trained on multiple profanity lists as well as the Wikidetox Toxic Comment Dataset (Wulczyn et al., 2016) and classifies a given string as being profane or not profane.

We considered only the English sentences in each dataset using the same language classifier from Section 3.7. We did this since `profanity-checker` is built for English and other languages may improperly impact the results. For instance, the German nominative/accusative feminine/plural definite article “die” is flagged as being profane regardless of context. We split each sentence into words and computed the percentage of words that are flagged as profane for each component of the Pile. We emphasize that this methodology is only a proxy for profanity, given the complexity of determining whether a given word or phrase is profane in context.

As shown in Figure 8, the Pile as a whole appears less profane than Pile-CC. Further, the majority of Pile components appear less profane than Pile-CC as well.

We also broke each dataset down on a sentence level, to allow `profanity-checker` to check entire sentences. Splitting datasets by sentence allows for additional context to be considered when determining whether content is pejorative. Our results are shown in Figure 12.

6.4 Bias and Sentiment Co-occurrence

As language models may pick up unexpected biases from the training data, we performed a preliminary analysis of the different components that make up the Pile. Because models with different characteristics may be trained on the Pile, we aimed to document the biases of the data and not a specific

model. We primarily focus on co-occurrence tests, where we analyzed what words occur in the same sentence as other specific words. Using this information, we can estimate what words strongly bias towards a category word, as well as calculate the general sentiment of surrounding words.

We focused our analysis on gender, religion, and race. Our goal is to provide users of this dataset with preliminary guidance on how the different components are biased so that they can make decisions on which components to train on.

All tables and figures in this section can be found in the Appendix.

6.4.1 Gender

We computed gender associations by computing co-occurrences for binary pronouns. For each word, we computed the difference in the rate it co-occurs with “he” and “she”¹⁴ and weighed it by the square root of its frequency. We report the top 15 most biased adjectives or adverbs (Loper and Bird, 2002) for each in Table 10. We see that words like “military”, “criminal”, and “offensive” strongly bias towards men, while “little”, “married”, “sexual”, and “happy” bias towards women.

In addition, we computed the average sentiment (Baccianella et al., 2010) of words co-occurring with the gendered pronouns across each dataset in Figure 13. Generally, we find no significant sentiment bias towards men or women. This, of course, does not mean that the dataset is free of gender bias (as our co-occurrence tests show).

6.4.2 Religion

We computed a similar co-occurrence analysis for religion, which can be found in Table 11. Like gender, we find that these co-occurrences reflect how these terms are used in pockets of online discourse. For example, “radical” co-occurs with “muslim” at a high rate, while “rational” often co-occurs with “atheist”. This analysis also demonstrates some of the limitations of a purely co-occurrence based analysis. For example, “religious” often co-occurs with “atheist”, which likely reflects the type of conversations in which the word “atheist” is likely to occur as opposed to a descriptor of “atheist”.

¹⁴We chose to only study male and female pronouns as a simplifying assumption. Studying “they” would require us to isolate its usage as a singular noun.

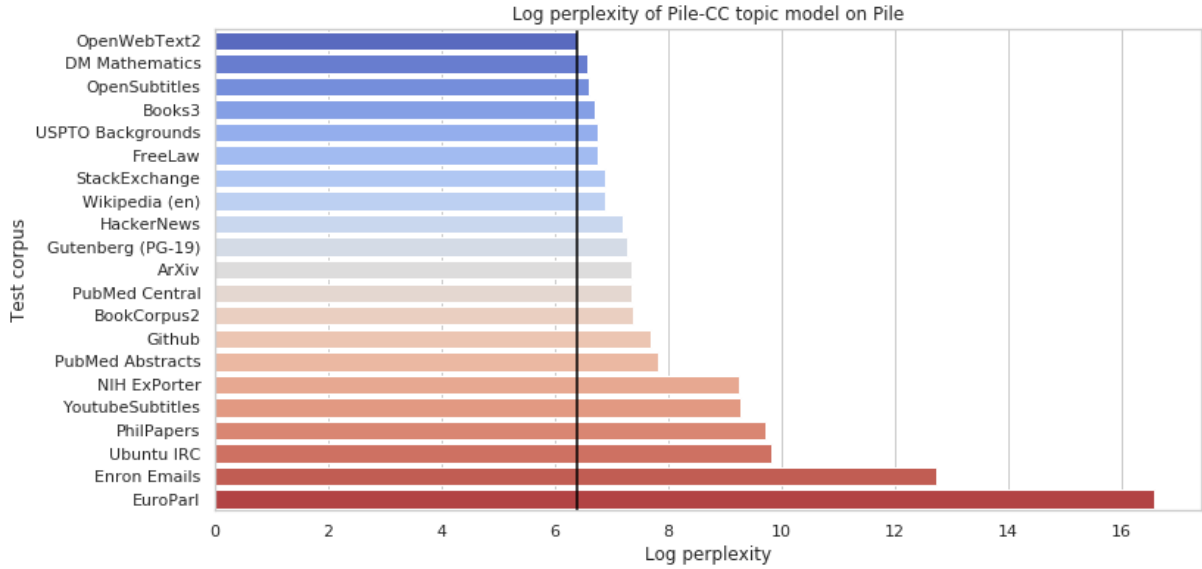


Figure 7: Log perplexity of 16-topic LDA trained on Pile-CC, on other Pile components. Dotted line indicates log perplexity of the topic model on OpenWebText2. Higher indicates a larger topical divergence from Pile-CC.

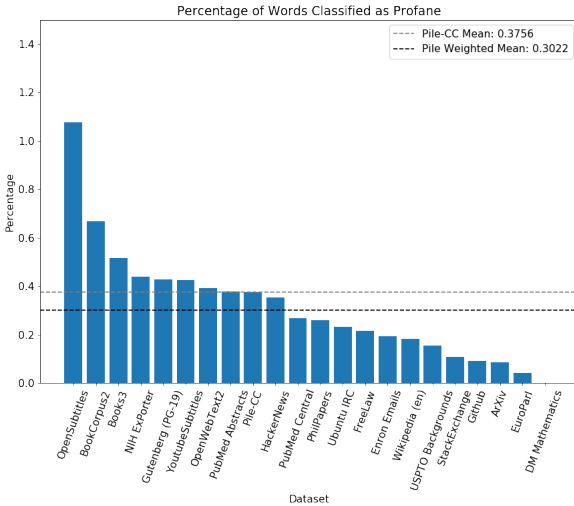


Figure 8: Percentage of words classified as profane in the Pile. The percentage of the CC component and the weighted mean of the Pile as a whole are shown as horizontal lines.

In addition, we computed the average sentiment of co-occurrences across each of the constituent datasets in Figure 14. Over the entire dataset, we find that “Buddhist” has the highest sentiment, followed by “Hindu”, “Christian”, “Atheist”, and “Muslim”. Notably, “Jew” is the lowest, perhaps reflecting its historical use as a pejorative.

6.4.3 Race

Finally, we ran the same analysis for racial groups. Here, as identifiers like “black” or “white” often do not indicate race, we instead compute co-

occurrences with phrases like “black man” or “white woman”.

We show the top 15 most biased words for each demographic in Table 12. Once again, we found that the co-occurrences reflect the context in which these terms are used. For example, the 4 most biased words for “black” are “unarmed”, “civil”, “criminal”, and “scary”.

Similar to above, we compute the average sentiment of co-occurring words. We report the average sentiment numbers in Table 13. We find that “hispanic/latino” narrowly edges out “asian” for the highest sentiment, followed by “white”. On the other hand, “black” had the lowest sentiment, at -0.15.

We note that for all demographics, the average sentiment is negative. We hypothesize that this is due to the specific context for which the phrases we use to compute co-occurrences appear. For example, it is often quite common for news articles to describe suspects as an “asian man”.

6.5 Author Consent and Public Data

Another issue with the use of texts in natural language processing research is consent. Although one is typically not legally obligated to receive the permission of an author to train a NLP algorithm on their work¹⁵, many consider doing so a moral obli-

¹⁵Laws vary by country. For a discussion of US law, see Section 7.1

gation or a good measure to guard against misuse (Obar, 2020; Prabhu and Birhane, 2020). On the other hand, there is significant disagreement surrounding the ethics of repurposing data protected by *terms of service* in research contexts (Vítak et al., 2016; Fiesler et al., 2020), particularly given the power asymmetries inherent in digital platforms, which often close off independent researchers from investigating public data while simultaneously compelling users to consent to its private use (Halavais, 2019).

While much of the Pile’s data comes from sources that have expressly consented to its wider dissemination and use in research, researchers often fail to clearly document where their data came from and under what terms its use was consented to. In light of this, we felt it appropriate to release the Pile with transparency around how the authors of its data have indicated that that data can be used.

To provide needed nuance to our discussion of consent, we identified three tiers of availability for public use. **Public data** is data which is freely and readily available on the internet. This primarily excludes data which is pay-walled (regardless of how easy that paywall is to bypass) and data which cannot be easily obtained but can be obtained, e.g. through a torrent or on the dark web. **Terms of Service (ToS) compliant data** is data which is obtained and used in a fashion that is known to be consistent with the terms of service of the data host. **Data with authorial consent** is data for which the original authors of the work consented to the use of their data, or where a reasonable person could not assume that their data would not be used for purposes such as research. ToS compliant data and authorial consented data differ in two main ways: It is important to keep in mind that people typically do not read Terms of Service, and additionally that being ToS-compliant does not entail authorial consent. We adopted a strict model of consent, where ambiguous or unknown consent is treated as non-consensual.

Table 5 summarizes our understanding of the status of each of the datasets within the Pile. Datasets marked with a ✓ are compliant in the relevant respects, though a couple datasets are worth remarking on in particular. Book3 and OpenSubtitles are being used in a fashion that is consistent with the terms of service of the data host. However, this is somewhat misleading in that the data host is not

authorized to post the data online by the parties that own it. The Enron Emails dataset was not collected with the permission of the authors, but was collected by the U.S. government as part of a criminal investigation. While the people whose emails are in the Enron dataset are aware of this fact, they were not given the ability to consent to its inclusion in any way.

There are five datasets included in the Pile that were not collected and distributed in a ToS compliant fashion and for which the authors had no ability to consent to their data being used. Each of these datasets are widely used, both in the NLP literature and the world at large. With the exception of the YouTube Subtitles dataset, each of these datasets were published by researchers and are passed around freely on the internet. The YouTube Subtitles dataset was created by us for this project, using a very popular unofficial API that is both widely used and easily obtainable on Pip, Conda, and GitHub, among other places. Given the processing applied and the difficulty of identifying particular files in the Pile, we feel that our use of these datasets does not constitute significantly increased harm beyond that which has already been done by the widespread publication of these datasets.

7 Implications and Broader Impacts

The Pile represents yet another stepping stone along the path of scaling models and datasets to ever larger sizes and capabilities. There are many serious concerns about how the emergence of progressively stronger AI systems will influence the wider world (Brundage et al., 2018; Amodei et al., 2016; Bostrom and Yudkowsky, 2014; Bostrom, 2014; Critch and Krueger, 2020), and we believe that they merit serious thought. In this section we discuss the legal ramifications of the Pile, and then consider the impact of the Pile to AI alignment from two angles: accelerating AI timelines and the dangers posed by unaligned language models.

7.1 Legality of Content

While the machine learning community has begun to discuss the issue of the legality of training models on copyright data, there is little acknowledgment of the fact that the processing and distribution of data owned by others may also be a violation of copyright law. As a step in that direc-

Component	Public	ToS	Author
Pile-CC	✓	✓	
PMC	✓	✓	✓
Books3	✓		
OWT2	✓		
ArXiv	✓	✓	✓
Github	✓	✓	
FreeLaw	✓	✓	✓
Stack Exchange	✓	✓	✓
USPTO	✓	✓	✓
PubMed	✓	✓	✓
PG-19	✓	✓	
OpenSubtitles	✓		
Wikipedia	✓	✓	✓
DM Math	✓	✓	✓
Ubuntu IRC	✓	✓	✓
BookCorpus2	✓		
EuroParl	✓	✓	✓
HackerNews	✓	✓	
YTSubtitles	✓		
PhilPapers	✓	✓	✓
NIH	✓	✓	✓
Enron Emails	✓	✓	

Table 5: Types of consent for each dataset

tion, we discuss the reasons we believe that our use of copyright data is in compliance with US copyright law.¹⁶

Under [pre \(1984\)](#) (and affirmed in subsequent rulings such as [aff \(2013\)](#); [Google \(2015\)](#)), non-commercial, not-for-profit use of copyright media is preemptively fair use. Additionally, our use is *transformative*, in the sense that the original form of the data is ineffective for our purposes and our form of the data is ineffective for the purposes of the original documents. Although we use the full text of copyright works, this is not necessarily disqualifying when the full work is necessary ([ful, 2003](#)). In our case, the long-term dependencies in natural language require that the full text be used in order to produce the best results ([Dai et al., 2019](#); [Rae et al., 2019](#); [Henighan et al., 2020](#); [Liu et al., 2018](#)).

Copyright law varies by country, and there may be

¹⁶This discussion does not, and is not intended to, constitute legal advice; rather, it is a general discussion of law. Only your attorney can provide assurances that the information contained herein is applicable or appropriate to a particular situation. If in doubt, it is always advisable to speak to an intellectual property attorney.

additional restrictions on some of these works in particular jurisdictions. To enable easier compliance with local laws, the Pile reproduction code is available and can be used to exclude certain components of the Pile which are inappropriate for the user. Unfortunately, we do not have the metadata necessary to determine exactly which texts are copyrighted, and so this can only be undertaken at the component level. Thus, this should be taken to be a heuristic rather than a precise determination.

7.2 Acceleration of AI Timelines

There is serious concern that AI systems may soon be meaningfully more capable than humans in all relevant economic tasks ([Grace et al., 2018](#); [Yudkowsky, 2013](#)). Relatedly, there are serious unresolved questions surrounding how to properly align such powerful AI systems with human interests ([Bostrom and Yudkowsky, 2014](#); [Russell, 2019](#); [Bostrom, 2014](#); [Amodei et al., 2016](#)) and generally avoid morally catastrophic outcomes ([Sotala and Gloor, 2017](#); [Shulman and Bostrom, 2020](#)). As such, it has been argued that accelerating the development of such powerful AI systems may be undesirable before these concerns have been more adequately addressed ([Bostrom, 2014](#)).

There are several pragmatic responses to this view:

1. Due to human competition, curiosity, and cultural diversity, halting technological development is incredibly difficult, if not impossible. ([Russell, 2019](#)) ([Critch and Krueger, 2020](#))
2. AI development is experimental in nature: The alignment problem can only be solved through development, testing and (hopefully non-existential) failure.
3. High powered language models, along with their more general successors, must be capable of viewing morally problematic content without adopting it in their output. We elaborate on this in the following section.

With this in mind, we accept the reality that the Pile could potentially accelerate AI timelines. However, we hope our efforts to establish best practices, such as thoroughly documenting the contents of our data, will help encourage diligence for downstream researchers on alignment problems.

7.3 Negative LM Output

There has been much discussion about the possible negative effects of powerful language models in the world (Brown et al., 2020; Brundage et al., 2018). Some of these possible problems, such as the ability to mass produce low quality content for the purpose of Search Engine Optimization, are inherent problems to the way online content is distributed, and cannot be stopped by those developing language models alone. Directly solving these problems would require sweeping changes to the architecture of the Internet, such as vastly expanded Public Key Infrastructure and distributed authentication of identity (Ferguson and Schneier, 2003).

Another concern is that training such models on huge datasets will almost inevitably require them to have undesirable content in their training sets, such as that promoting hateful stereotypes (Christian, 2020). Having models output undesirable content is, by definition, undesirable, but we believe that attacking this problem from the training set side is unproductive and ultimately leads us away from optimal solutions. If a person reads a racist piece of content, they do not then immediately adopt its racist views—they may be capable of doing so, but can decide not to. This capacity to understand undesirable content and then decide to ignore it is an essential future research direction. Not only would this allow models to use “dirtier” data with less concern, but also to use their gained knowledge to better understand what not to do. We recognize that, despite recent progress in human-guided learning (Stiennon et al., 2020), the technology is not yet at this stage, and have thus made a number of editorial decisions as described in this paper. However, this approach seems essential to the future of these models and AI more broadly, and more research is needed.

8 Related Work

Self-supervised training of natural language processing models on large, unlabeled text corpora, has seen widespread adoption in the field. Word representation models such as GloVe (Pennington et al., 2014) and word2vec (Mikolov et al., 2013) were trained on datasets such as Wikipedia, Gigaword (Graff et al., 2003), or a non-public Google News corpus. More recently, language models (Radford et al., 2018, 2019; Brown et al., 2020;

Rosset, 2019; Shueybi et al., 2019) and masked language models (Devlin et al., 2019; Liu et al., 2019; Raffel et al., 2019) have been trained on datasets such as Wikipedia, BookCorpus (Zhu et al., 2015), RealNews (Zellers et al., 2019), CC-Stories (Trinh and Le, 2018), and other Internet scrape-derived datasets discussed below. Other datasets such as WikiText (Stephen et al., 2016) have also been used in similar self-supervised training.

As data requirements for language modeling have grown, the field has turned towards Internet scrapes for large-scale datasets (Gokaslan and Cohen, 2019), with Common Crawl being particularly prevalent. Works such as Brown et al. (2020); Wenzek et al. (2019); Suárez et al. (2019b); Raffel et al. (2019) have relied on Common Crawl to build training datasets for large-scale models. However, these works often highlight the difficulty of cleaning and filtering the Common Crawl data, and often highlight the resulting data quality as a determining factor of model capability.

It has also been increasingly common practice to combine multiple datasets when training language models. For instance, GPT (Radford et al., 2018) was trained on Wikipedia and BookCorpus, whereas GPT-3 (Brown et al., 2020) was trained on Wikipedia, two fiction datasets, and two web-scraped datasets. The Pile continues the trend of combining large-scale web-scrapes with smaller, higher-quality datasets that capture knowledge we believe would be most beneficial to training language models.

The two most comparable publicly available datasets to the Pile are CC-100 (Wenzek et al., 2019) and C4/mC4 (Raffel et al., 2019). C4 is comparably-sized to the Pile, while mC4 and CC-100 are larger, multilingual datasets. However, C4/mC4 require immense computational resources to preprocess the data, with its maintainers even recommending the use of a distributed cloud service,¹⁷ setting a high bar of entry to using these datasets. CC-100 is directly downloadable and pre-cleaned; however, its English portion is much smaller than the Pile. Importantly, these three datasets are all derived entirely from Common Crawl—as discussed above, the current best practice in training large-scale language models involve using both large web scrapes and more targeted, higher-quality datasets,

¹⁷<https://www.tensorflow.org/datasets/catalog/c4>

which the Pile directly addresses.

9 Acknowledgments

The authors would like to thank TensorFlow Research Cloud for providing the computational resources for the evaluation and OpenAI for providing access and credits for the OpenAI API for GPT-3 evaluation.

We would also like to thank Farrukh Raman, JR Smith, and Michelle Schmitz for reviewing the manuscript.

References

1984. Sony corp. of america v. universal city studios, inc.
2003. Kelly v. arriba soft corp.
2013. Righthaven llc v. hoehn.
- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. 2016. [Concrete problems in AI safety](#). *arXiv preprint arXiv:1606.06565*.
- Stefano Baccianella, Andrea Esuli, and Fabrizio Sebastiani. 2010. [Sentiwordnet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining](#). In *LREC*. European Language Resources Association.
- Emily M Bender and Batya Friedman. 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.
- Stella Biderman. 2021. Data statement for the Pile. *arXiv preprint arXiv*.
- Stella Biderman, Kieran Bicheno, and Leo Gao. 2021. Datasheet for the Pile. *arXiv preprint arXiv*.
- Stella Biderman and Walter J. Scheirer. 2020. Pitfalls in machine learning research: Reexamining the development cycle. *NeurIPS “I Can’t Believe It’s Not Better!” Workshop*.
- David M Blei, Andrew Y Ng, and Michael I Jordan. 2003. Latent dirichlet allocation. *Journal of machine Learning research*, 3(Jan):993–1022.
- Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. Language (technology) is power: A critical survey of “bias” in NLP. *arXiv preprint arXiv:2005.14050*.
- Nick Bostrom. 2014. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, Inc.
- Nick Bostrom and Eliezer Yudkowsky. 2014. The ethics of artificial intelligence. *The Cambridge handbook of artificial intelligence*, 1:316–334.
- Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Nee-lakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*.
- Miles Brundage, Shahar Avin, Jack Clark, Helen Toner, Peter Eckersley, Ben Garfinkel, Allan Dafoe, Paul Scharre, Thomas Zeitzoff, Bobby Filar, et al. 2018. The malicious use of artificial intelligence: Forecasting, prevention, and mitigation. *arXiv preprint arXiv:1802.07228*.
- Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlings-son, Alina Oprea, and Colin Raffel. 2020. [Extracting training data from large language models](#).
- Isaac Caswell, Theresa Breiner, Daan van Esch, and Ankur Bapna. 2020. Language id in the wild: Unexpected challenges on the path to a thousand-language web text corpus. *arXiv preprint arXiv:2010.14571*.
- Brian Christian. 2020. *The Alignment Problem: Machine Learning and Human Values*. WW Norton & Company.
- Alina Maria Ciobanu, Liviu P Dinu, and Andrea Sgarro. 2017. Towards a map of the syntactic similarity of languages. In *International Conference on Computational Linguistics and Intelligent Text Processing*, pages 576–590. Springer.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.
- Marta R Costa-jussà, Roger Creus, Oriol Domingo, Albert Domínguez, Miquel Escobar, Cayetana López, Marina García, and Margarita Geleta. 2020. Mt-adapted datasheets for datasets: Template and repository. *arXiv preprint arXiv:2005.13156*.
- Andrew Critch and David Krueger. 2020. AI Research Considerations for Human Existential Safety (ARCHES). *Preprint at [acritch.com/arches](#)*.
- Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V Le, and Ruslan Salakhutdinov. 2019. Transformer-xl: Attentive language models beyond a fixed-length context. *arXiv preprint arXiv:1901.02860*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics.

- István Endrédy and Attila Novák. 2013. More effective boilerplate removal – the GoldMiner algorithm. In *Polibits*.
- Niels Ferguson and Bruce Schneier. 2003. *Practical Cryptography*. John Wiley & Sons.
- Casey Fiesler, Nathan Beard, and Brian C Keegan. 2020. No robots, spiders, or scrapers: Legal and ethical regulation of data collection methods in social media terms of service. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 14, pages 187–196.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. 2018. Datasheets for datasets. *arXiv preprint arXiv:1803.09010*.
- Aaron Gokaslan and Vanya Cohen. 2019. Openweb-text corpus. <http://Skylion007.github.io/OpenWebTextCorpus>.
- Authors Guild v. Google. 2015. . *Docket No. 13-4829-cv*, 804:202.
- Katja Grace, John Salvatier, Allan Dafoe, Baobao Zhang, and Owain Evans. 2018. When will AI exceed human performance? evidence from AI experts. *Journal of Artificial Intelligence Research*, 62:729–754.
- David Graff, Junbo Kong, Ke Chen, and Kazuaki Maeda. 2003. English gigaword. *Linguistic Data Consortium, Philadelphia*, 4(1):34.
- Declan Groves and Andy Way. 2006. Hybridity in mt: Experiments on the Europarl corpus. In *Proceedings of the 11th Annual conference of the European Association for Machine Translation (EAMT 2006)*.
- Alexander Halavais. 2019. Overcoming terms of service: a proposal for ethical distributed research. *Information, Communication & Society*, 22(11):1567–1581.
- Chris Hardin. 2018. [How to shuffle a big dataset](#).
- Tom Henighan, Jared Kaplan, Mor Katz, Mark Chen, Christopher Hesse, Jacob Jackson, Heewoo Jun, Tom B Brown, Prafulla Dhariwal, Scott Gray, et al. 2020. Scaling laws for autoregressive generative modeling. *arXiv preprint arXiv:2010.14701*.
- Matthew Hoffman, Francis Bach, and David Blei. 2010. Online learning for latent dirichlet allocation. *advances in neural information processing systems*, 23:856–864.
- Dirk Hovy and Shannon L Spruit. 2016. The social impact of natural language processing. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 591–598.
- Ben Hutchinson, Vinodkumar Prabhakaran, Emily Denton, Kellie Webster, Yu Zhong, and Stephen Denyul. 2020. Social biases in NLP models as barriers for persons with disabilities. *arXiv preprint arXiv:2005.00813*.
- Eun Seo Jo and Timnit Gebru. 2020. Lessons from archives: Strategies for collecting sociocultural data in machine learning. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pages 306–316.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- Bryan Klimt and Yiming Yang. 2004. The Enron corpus: A new dataset for email classification research. In *European Conference on Machine Learning*, pages 217–226. Springer.
- Sosuke Kobayashi. 2018. Homemade bookcorpus. <https://github.com/BIGBALLON/cifar-10-cnn>.
- Philipp Koehn. 2005. Europarl: A parallel corpus for statistical machine translation. In *MT summit*, volume 5, pages 79–86. Citeseer.
- Dmitry Lepikhin, HyounJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, and Zhifeng Chen. 2020. Gshard: Scaling giant models with conditional computation and automatic sharding. *arXiv preprint arXiv:2006.16668*.
- Peter J Liu, Mohammad Saleh, Etienne Pot, Ben Goodrich, Ryan Sepassi, Lukasz Kaiser, and Noam Shazeer. 2018. Generating wikipedia by summarizing long sequences. *arXiv preprint arXiv:1801.10198*.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Edward Loper and Steven Bird. 2002. NLTK: The Natural Language Toolkit. In *Proceedings of the ACL Workshop on Effective Tools and Methodologies for Teaching Natural Language Processing and Computational Linguistics*, pages 62–69. Somerset, NJ: Association for Computational Linguistics. <http://arXiv.org/abs/cs/0205028>.
- John MacFarlane. 2006–2020. [Pandoc: a universal document converter](#).
- Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013. [Distributed representations of words and phrases and their compositionality](#). In *Advances in Neural Information Processing Systems*, volume 26, pages 3111–3119. Curran Associates, Inc.
- Jonathan A Obar. 2020. Sunlight alone is not a disinfectant: Consent and the futility of opening big data black boxes (without assistance). *Big Data & Society*, 7(1):2053951720935615.
- Jeffrey Pennington, Richard Socher, and Christopher D. Manning. 2014. [Glove: Global vectors for word repre-](#)

- sentation. In *Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543.
- Vinay Uday Prabhu and Abeba Birhane. 2020. Large image datasets: A pyrrhic win for computer vision? *arXiv preprint arXiv:2006.16923*.
- Shawn Presser. 2020. Books3. <https://twitter.com/theshawwn/status/1320282149329784833>.
- Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. 2018. Improving language understanding with unsupervised learning. *Technical report, OpenAI*.
- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *OpenAI Blog*, 1(8):9.
- Jack W Rae, Anna Potapenko, Siddhant M Jayakumar, Chloe Hillier, and Timothy P Lillicrap. 2019. *Compressive transformers for long-range sequence modelling*. *arXiv preprint*.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv preprint arXiv:1910.10683*.
- Inioluwa Deborah Raji and Jingying Yang. 2019. ABOUT ML: Annotation and benchmarking on understanding and transparency of machine learning lifecycles. *arXiv preprint arXiv:1912.06166*.
- C. Radhakrishna Rao. 1961. *Generation of random permutations of given number of elements using random sampling numbers*. *Sankhyā: The Indian Journal of Statistics, Series A (1961-2002)*, 23(3):305–307.
- Radim Rehurek, Petr Sojka, et al. 2011. Gensim—statistical semantics in python. *NLP Centre, Faculty of Informatics, Masaryk University*.
- C Rosset. 2019. Turing-NLG: A 17-billion-parameter language model by Microsoft. *Microsoft Blog*.
- S. Russell. 2019. *Human Compatible: Artificial Intelligence and the Problem of Control*. Penguin Publishing Group.
- David Saxton, Edward Grefenstette, Felix Hill, and Pushmeet Kohli. 2019. Analysing mathematical reasoning abilities of neural models. *arXiv preprint arXiv:1904.01557*.
- Ismaïla Seck, Khoulood Dahmane, Pierre Duthon, and Gaëlle Loosli. 2018. Baselines and a datasheet for the Cerema AWP dataset. *arXiv preprint arXiv:1806.04016*.
- Mohammad Shoeybi, Mostofa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. 2019. Megatron-LM: Training multi-billion parameter language models using gpu model parallelism. *arXiv preprint arXiv:1909.08053*.
- Carl Shulman and Nick Bostrom. 2020. Sharing the world with digital minds. *preprint*.
- Kaj Sotala and Lukas Gloor. 2017. Superintelligence as a cause or cure for risks of astronomical suffering. *Informatica*, 41(4).
- Robyn Speer. 2019. *ftfy*. Zenodo. Version 5.5.
- Merity Stephen, Xiong Caiming, Bradbury James, and Richard Socher. 2016.
- Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F Christiano. 2020. Learning to summarize with human feedback. *Advances in Neural Information Processing Systems*, 33.
- Pedro Javier Ortiz Suárez, Benoît Sagot, and Laurent Romary. 2019a. Asynchronous pipeline for processing huge corpora on medium to low resource infrastructures. In *7th Workshop on the Challenges in the Management of Large Corpora (CMLC-7)*. Leibniz-Institut für Deutsche Sprache.
- Pedro Javier Ortiz Suárez, Benoît Sagot, and Laurent Romary. 2019b. Asynchronous pipeline for processing huge corpora on medium to low resource infrastructures. In *7th Workshop on the Challenges in the Management of Large Corpora (CMLC-7)*. Leibniz-Institut für Deutsche Sprache.
- Anja Thieme, Danielle Belgrave, and Gavin Doherty. 2020. Machine learning in mental health: A systematic review of the HCI literature to support the development of effective and implementable ML systems. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 27(5):1–53.
- J. Tiedemann. 2016. Finding alternative translations in a large corpus of movie subtitles. *Proceedings of the 10th International Conference on Language Resources and Evaluation (LREC 2016)*.
- Trieu H. Trinh and Quoc V. Le. 2018. *A simple method for commonsense reasoning*. *CoRR*, abs/1806.02847.
- Hans Van Halteren. 2008. Source language markers in Europarl translations. In *Proceedings of the 22nd International Conference on Computational Linguistics (Coling 2008)*, pages 937–944.
- Jessica Vitak, Katie Shilton, and Zahra Ashktorab. 2016. Beyond the Belmont principles: Ethical challenges, practices, and beliefs in the online data research community. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*, pages 941–953.
- Guillaume Wenzek, Marie-Anne Lachaux, Alexis Conneau, Vishrav Chaudhary, Francisco Guzmán, Armand Joulin, and Edouard Grave. 2019. CCNet: Extracting high quality monolingual datasets from web crawl data. *arXiv preprint arXiv:1911.00359*.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe

Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Ellery Wulczyn, Nithum Thain, and Lucas Dixon. 2016. [Wikipedia detox](#).

Eliezer Yudkowsky. 2013. Intelligence explosion microeconomics. *Machine Intelligence Research Institute*, accessed online October, 23:2015.

Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. [Defending against neural fake news](#). In H. Wallach, H. Larochelle, A. Beygelzimer, F. d’Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems 32*, pages 9054–9065. Curran Associates, Inc.

Victor Zhou. 2019. Building a better profanity detection library with scikit-learn.

Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. 2015. Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *Proceedings of the IEEE international conference on computer vision*, pages 19–27.